

ASHWIN GUDURU

Data Engineer

Ashwinguduru630@gmail.com | +1 630 635 7962 | [Linkedin](#)

SUMMARY

Experienced Data Engineer with 4 years of expertise in Python, R, SQL, and PostgreSQL, utilizing IDEs like PyCharm, Jupyter Notebook, and Visual Studio Code. Proficient in big data technologies (HDFS, Hadoop, Spark, Kafka, Apache Airflow), ETL/ELT pipeline development, and cloud platforms (AWS, Azure). Skilled in database management (PostgreSQL, MySQL, MongoDB, Oracle, Aurora, Redshift, Snowflake) and experience in data governance, data modeling, and job scheduling using Apache Airflow. Well-versed in DevOps practices (GitHub, GitLab, CI/CD) and containerization tools like Docker and Kubernetes. Strong background in data analysis using packages (NumPy, Pandas, TensorFlow) and proficient in data Visualization using Tableau and Power BI. Experienced with Alteryx, Informatica, and Erwin Data Modeler. Comfortable Working with MS Office, and operating systems (Windows, Linux).

EXPERIENCE

Data Engineer | Charles Schwab, IL

January 2024 - Present

- **ETL/ELT Pipeline Development & Automation:** planned and automated **ETL/ELT pipelines** using **Python, AWS Glue, Lambda, and Apache Airflow** for data ingestion, transformation, and scheduling. Streamlined workflows into **PostgreSQL and Amazon Redshift**, reducing manual handling by **80%** and improving efficiency by **60%** through computerized monitoring.
- **Cloud Data Warehouse Implementation:** Deployed and optimized a **Snowflake-based data warehouse** integrated with **AWS (S3, Glue, Lambda)** for real-time and batch processing, enhancing reporting speed by **35%**.
- **Machine Learning Integration for Predictive Analytics:** Integrated machine learning models using **Python, TensorFlow**, and **Scikit-learn** into existing data pipelines to predict customer behaviour, which led to a **20%** increase in retention rates.
- **Multi-Environment Development & Deployment:** Developed in **multi-environment setups (Dev, QA, Prod)** and established **CI/CD pipelines (GitLab, Terraform, Jenkins)** for systematized testing and deployment, reducing deployment time by **40%**.
- **Docker-based Containerization:** Developed and deployed containerized **ETL workflows** using **Docker** and **Kubernetes**, reducing deployment errors by **35%** and improving deployment time by **50%**.
- **Job Scheduling & Workflow Automation:** created and managed **Apache Airflow DAGS** to orchestrate and schedule complex data workflows across multiple systems. Ensured **99.9%** uptime in data processing with error handling and monitoring mechanisms.
- **Real-time Data Processing:** Implemented **AWS Kinesis and Apache Kafka** for real-time data ingestion and streaming analytics. Enhanced **fraud detection and anomaly detection accuracy by 20%**, enabling faster response to high-risk transactions.

Data Engineer | Bytecraft System, IND

July 2019 - July 2022

- **Data Lake & NoSQL Implementation:** Built a scalable data lake on **Azure Blob Storage and HDFS**, integrating **Hive for structured data** and **MongoDB (vNoSQL) for unstructured data** like doctor's notes and sensor logs, improving analytics by **45%**.
- **ETL & Data Transformation Automation:** developed and mechanized **ETL pipelines** using **Informatica, Azure Data Factory, and Python** for Electronic Health Records (**EHR**) processing, claims data integration, and patient demographics, reducing manual effort by **40%**.
- **Data Model Optimization:** Optimized **clinical and financial reporting models** in **Azure Synapse and SQL Server**, improving report generation by **30%** for **patient outcomes, hospital operations, and revenue cycle management**.
- **Data Quality & Integrity Assurance:** Designed **data validation frameworks** in **Informatica, Python, and SQL** to ensure accuracy in **patient records, insurance claims, and clinical trials**, reducing data discrepancies by **98%**.
- **Data Visualization:** Built interactive dashboards and dynamic reports in **Tableau** and **Power BI** using **Azure Synapse and ADF**, tracking **hospital performance, patient readmission rates, and treatment effectiveness**, improving KPIs by **25%**.
- **CI/CD Implementation:** Developed **CI/CD pipelines** with **Terraform, Jenkins, and GitHub Actions** for **data integration, compliance reporting, and medical research**, ensuring **scalability, security, and HIPAA compliance**.

SKILLS

Programming Language: R, Python, SQL

IDE's: PyCharm, Jupyter Notebook, Visual Studio Code

Big Data Ecosystem: HDFS, Apache Hadoop, Apache Spark, Apache Kafka, Apache Airflow, Scala, Databricks, Hive, Pig

Cloud Platforms: AWS (EC2, S3, RDS, Lambda, Glue, Kinesis), Azure (Data Factory, Blob Storage, Synapse Analytics)

ETL Tools: Informatica, Data Modeling Erwin 3.5.2

DevOps & Deployment: Docker, Kubernetes, Github, GitLab, CI/CD (Jenkins, Terraform)

Data Visualization Tools: Tableau, Power BI, SAS, Qlik Sense

Database: Snowflake, SQL Databases (PostgreSQL, MySQL), NoSQL Databases (MongoDB, Cassandra)

PROJECTS

Marketing Campaign data analysis:

- Boosted marketing campaign ROI by 25% through a Python and SQL-based ETL pipeline that analyzed user demographics and product preferences. Improved query performance by 30% and optimized strategies by identifying key customer segments.

Real-time Fraud Detection in Financial Transactions

- Developed a real-time streaming pipeline with Apache Kafka and Spark to process financial transactions and detect anomalies, triggering alerts for suspicious activities.

EDUCATION

Master of Science (M.S.) in Business Analytics and Information Systems
University of South Florida (USF) | Tampa, Florida

May 2024